

your function should emit the following sub-groups:

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<"star", "rats">,
<"src", "car", "rac">,
<"cat", "tac">,
<"now", "won">,
<"new">,
<"not">.
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What is the time and space complexity of your code?

10. You are given a directed graph $G \langle V, E \rangle$ where V is the set of vertices and E is the set of edges. You are given two vertices $N1$ and $N2$. You are asked to write a program to calculate how many paths exist between $N1$ and $N2$. What is the time and space complexity of your code?
11. Consider the problem discussed in Question (10) above. Now you are asked to find out how many paths exist between each pair of vertices $\langle v1, v2 \rangle$ that are present in graph G . What is the time complexity of your code? Would you be able to reuse your solution from Question (10)? If you can do so, does it increase the complexity of your code? Explain why.
12. You are given an array of unsorted N integers. You are asked to write a program which can return the K th largest value and K th smallest value in the array, where K can be an arbitrary integer. What is the time and space complexity of your code? Can you do this without sorting the array?
13. You are given two lists of integers, $L1$ and $L2$. You are asked to write a program which can merge the two lists, $L1$ and $L2$, and create a list $L3$. What is the time complexity of your solution if the two lists $L1$ and $L2$ are unsorted lists of integers? What is the time complexity of your solution if the two lists $L1$ and $L2$ are already sorted? If you are now asked to do the merging of lists in place, what would be the time complexity for both the following cases: (a) unsorted lists, and (b) sorted lists?
14. Given an array of N integers, you are asked to write a program that checks whether the array is an arithmetic sequence. An arithmetic sequence is one in which there is a constant difference between two consecutive elements in the sequence. For example, the sequence $\langle 1, 5, 9, 13, 17, 21 \rangle$ forms an arithmetic sequence, whereas the sequence $\langle 1, 4, 7, 10, 12, 15, 18, 20 \rangle$ does not form an arithmetic sequence. What is the time complexity of your code?
15. You are given a large piece of text and asked to write a program that can compute the frequencies of different characters in the text. For instance, given the text, "I came to see you," the output should be: $\langle (I, 1), (c, 1), (a, 1), (m, 1), (e, 3), (t, 1), (o, 2), (t, 1), (y, 1), (u, 1) \rangle$. First assume that the text consists

of English language characters and is of fixed size. What is the time and space complexity of your solution? You are now told that the text can consist of any Unicode character. How would you modify your earlier solution to handle this?

16. Given an array of N integers, you are asked to write a program that can check whether the array can be split into two sub-arrays, $A1$ and $A2$, such that the sum of elements in $A1$ has the same value as the sum of elements in $A2$. For example, given the array A as $\langle 5, 10, 15, 20 \rangle$, it is possible to split it into two sub-arrays $A1 = \langle 5, 20 \rangle$ and $A2 = \langle 10, 15 \rangle$ whose sum of elements are equal.
 17. You are given a sorted list of N integers and asked to create a binary search tree. What is the time complexity of your solution? Can you explain whether the binary search tree that gets constructed is a unique solution? If yes, explain why it has to be unique. If not, give a counter-example.
 18. Given a directed unweighted graph $G(V, E)$, where V is the set of vertices and E is the set of edges, find a spanning tree for the graph. (Note that you are asked to find a spanning tree and not a minimum spanning tree.) Is the spanning tree you construct unique for a given graph? If yes, explain why. If not, give a counter-example.
 19. You are given two lists of integers, $L1$ and $L2$, such that $L1$ is of length N and $L2$ is of length M . N is very much longer than M . You are asked to find a sub-sequence in $L1$ that can cover all elements in $L2$. Note that the sub-sequence can also contain elements not present in $L2$. What is the time complexity of your solution?
 20. You are given an array of N integers. The elements of the array can either be positive or negative. Find all triples in the array such that their sum is zero.
- If you have any favourite programming questions/software topics that you would like to discuss on this forum, please send them to me, along with your solutions and feedback, at sandyasm_AT_yahoo_DOT_com. Till we meet again next month, happy programming! 

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